JAVA PROJECT DOCUMENTATION

NA Browser

A Web Browser Using JavaFX

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Chapter 1

Introduction

JavaFX is a Java-based framework used to develop graphical user interface applications. It provides components such as buttons, text fields, menus, and layouts.

NA Browser is a desktop web browser developed using Java and JavaFX. It uses the WebView and WebEngine classes to load and display websites within the application window.

Chapter 2

Objectives

The main objectives of the project are:

- To develop a simple and user-friendly desktop web browser.
- To understand JavaFX GUI development.
- To implement URL loading and web navigation.
- To provide Back, Forward, and Refresh functionality.
- To understand event handling in Java.

Chapter 3

Requirements

3.1 Hardware Requirements

- Computer or laptop.
- Minimum 4 GB RAM.
- Internet connection.

3.2 Software Requirements

- Java Development Kit (JDK).
- JavaFX libraries.
- IntelliJ IDEA, Eclipse, or Visual Studio Code.
- Apache Maven, if used for dependency management.
- Windows, Linux, or macOS.

Chapter 4

System Design

The browser consists of the following components:

- **User Interface:** Provides the address bar and navigation buttons.
- **WebView:** Displays web pages.
- **WebEngine:** Loads URLs and manages web navigation.
- **Event Handlers:** Respond to button clicks and user input.

4.1 Workflow

1. The user opens NA Browser.
2. The browser interface is displayed.
3. The user enters a website URL.
4. WebEngine loads the requested website.
5. WebView displays the web page.
6. The user navigates using the available controls.

Chapter 5

Implementation

5.1 Application Window

The Stage class creates the main application window, while the Scene class contains the user interface.

5.2 Web Page Display

The WebView class displays web content, and the WebEngine class loads and controls web pages.

5.3 Navigation Controls

Buttons are used to implement Back, Forward, and Refresh operations through event handlers.

5.4 URL Handling

A text field allows users to enter website addresses. The entered URL is passed to WebEngine to load the requested page.

5.5 Testing

The browser should be tested for website loading, URL input, navigation controls, and handling of invalid URLs.

Chapter 6

Conclusion

NA Browser demonstrates the development of a desktop application using Java and JavaFX. The project provides practical experience with GUI components, event handling, and web page integration.

Bibliography

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